Sisi Zhang

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Research Interests

Applied Econometrics, Energy Economics, Environmental Economics, Agricultural Economics, Climate Economics, Public Policy

Education

Ph.D., Economics — University of Hawaii at Manoa, 2024

- Dissertation: *Essays on Energy Demand, Agricultural Yield Forecasting, and Climate*
  - Chapter 1: Using Temperature Sensitivity to Estimate Shiftable Electricity Demand
  - Chapter 2: Comparing Process-Based SIMPLE Model and Statistical Model with Climate Implications for U.S. Corn Belt States
  - Chapter 3: Comparison of Reanalysis Weather Datasets (NARR, MERRA2, ERA5-Land) for Hourly Electricity Demand Forecasting in the Continental U.S.
- TK Jayaraman & Family Prize for Best Ph.D. Dissertation

M.A., Economics — University of Hawaii at Manoa, 2019

M.Sc., Risk Management — Duisenberg School of Finance, Amsterdam, 2012

B.Sc., Economics — Tilburg University, 2010 — Cum laude (top 5%)

Academic Appointments

Lecturer — University of Hawaii at Manoa, Department of Economics, January 2020 – present (full-time Fall 2024; part-time all other semesters)

Research Assistant — University of Hawaii at Manoa, Department of Economics, August 2018 – August 2023

Other Professional Appointments

Economist VI (Senior Economist) — State of Hawaii, Department of Taxation, January 2025 – present

Co-founder & CEO — AE4cast (Agricultural Yield Forecasting), 2023 – present

Research Analyst, Grain — Glencore Grain B.V., Rotterdam, Netherlands, October 2015 – July 2017

Risk Control Analyst — Shell Trading B.V., Rotterdam, Netherlands, August 2012 – April 2015

Publications

Roberts, M. J., **Zhang, S.**, Yuan, E., Jones, J., and Fripp, M. (2022). "Using Temperature Sensitivity to Estimate Shiftable Electricity Demand." *iScience*, 25(9), 104940. DOI: 10.1016/j.isci.2022.104940.

Working Papers

Zhang, S. (2026). "Process-Based Crop Models versus Statistical Models: Divergent Climate Change Projections for U.S. Corn and Soybean Yields"

Zhang, S. (2026). "Comparison of Reanalysis Weather Data for Hourly Electricity Demand Forecast"

Roberts, M. J. and **Zhang, S.** (2024). "Using Weather-Based Crop Yield Forecasts to Estimate Commodity Demand." Presented at J.P. Morgan Center for Commodities Symposium, August 2024.

Teaching Experience

University of Hawaii at Manoa, Department of Economics

Instructor of Record (January 2020 – present):

- **ECON 130: Introduction to Microeconomics** — In-person. Multiple semesters. Fall 2024 enrollment: 147 students (primarily first-year, mixed majors).
- **ECON 321: Introduction to Statistics** — In-person and online. Multiple semesters.
- **ECON 358: Environmental Economics** — Online asynchronous. Multiple semesters. Designed in both 16-week and 5-week accelerated formats.
- **ECON 420: Mathematical Economics** — Online. Multiple semesters.
- **ECON 425: Introduction to Econometrics** — Online asynchronous. Multiple semesters.

Early College Instructor (Fall 2024):

- ECON 130: Introduction to Microeconomics — Maryknoll School, through Windward Community College.

Online Course Design & Instructional Innovation

- Completed Quality Matters training. Designed online asynchronous courses for ECON 358 and ECON 425 that met UH Quality Online Standards, aligned with Quality Matters and SUNY OSCQR frameworks, in both 16-week and 5-week accelerated formats.
- Platforms: Laulima, Brightspace, Cengage MindTap, WebAssign; R programming assignments in econometrics.

Conference Presentations

- J.P. Morgan Center for Commodities Symposium, August 2024 — "Using Weather-Based Crop Yield Forecasts to Estimate Commodity Demand" (with M. J. Roberts)
- Western Economic Association International (WEAI), 2021
- U.S. Association for Energy Economics (USAEE), 2021

Technical Skills & Languages

Programming: R, Python, Stata, SQL, MATLAB, SAS, EViews, SPSS, VBA

Data & Tools: Tableau, Google Earth Engine, ENVI, Git, LaTeX, Microsoft Office

Methods: Causal inference, panel data econometrics, time-series forecasting, machine learning, demand estimation, cross-validation, policy impact evaluation

Languages: Mandarin Chinese (native), English (professional fluency)